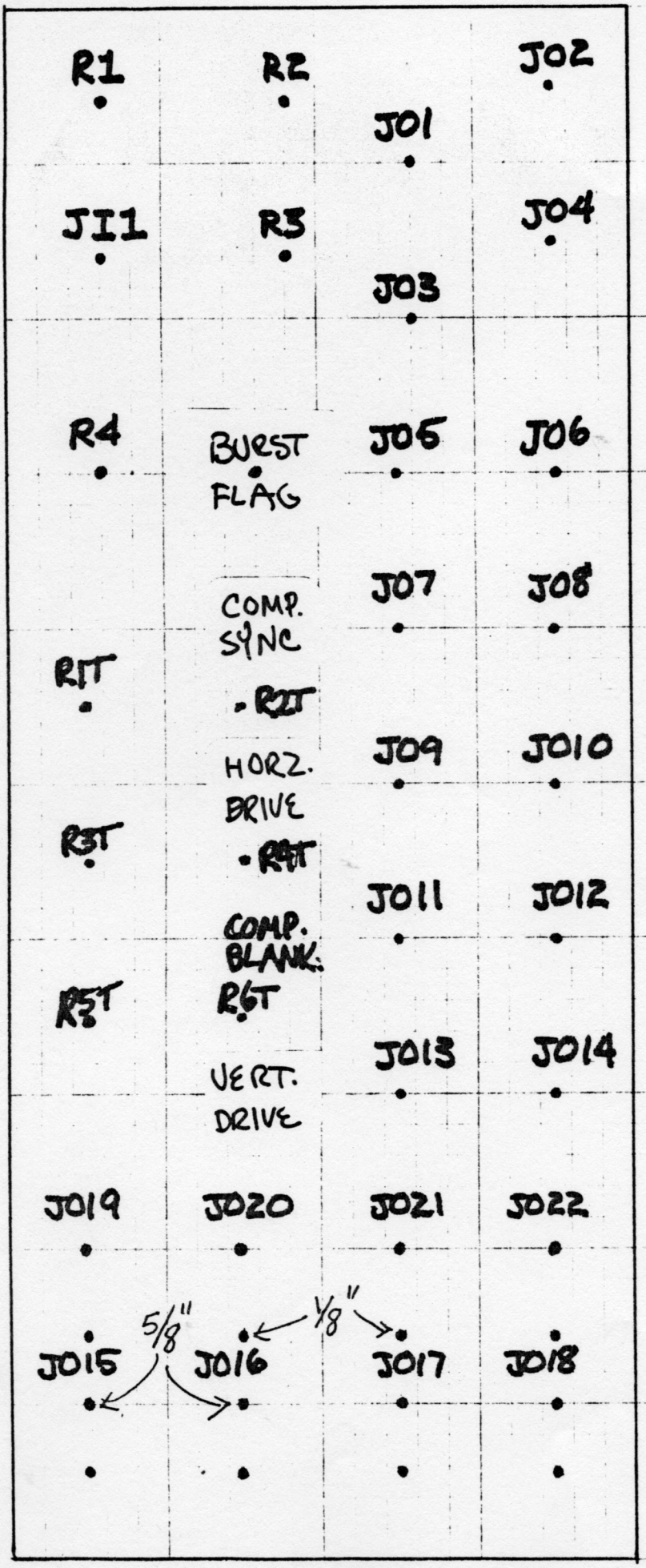
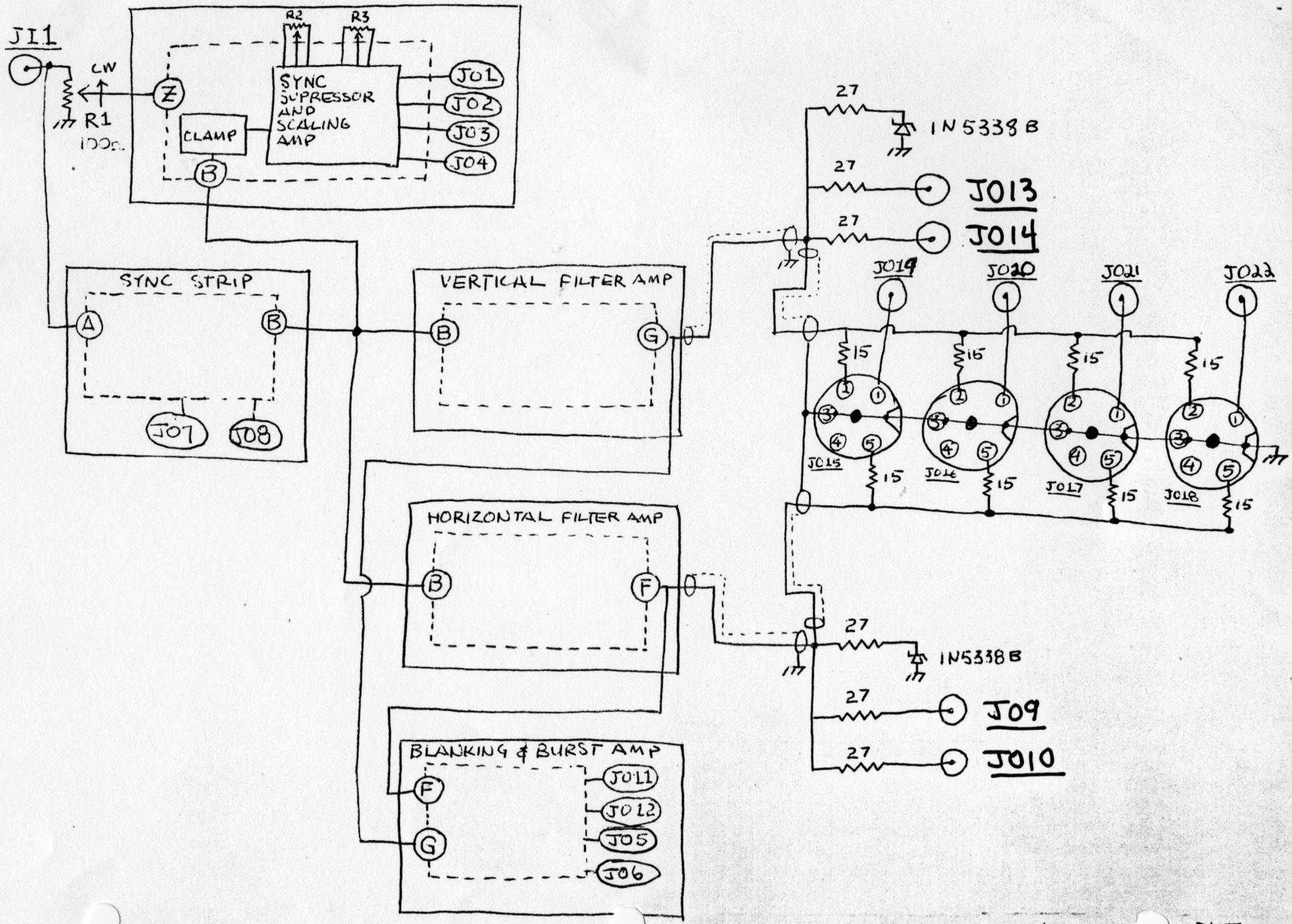
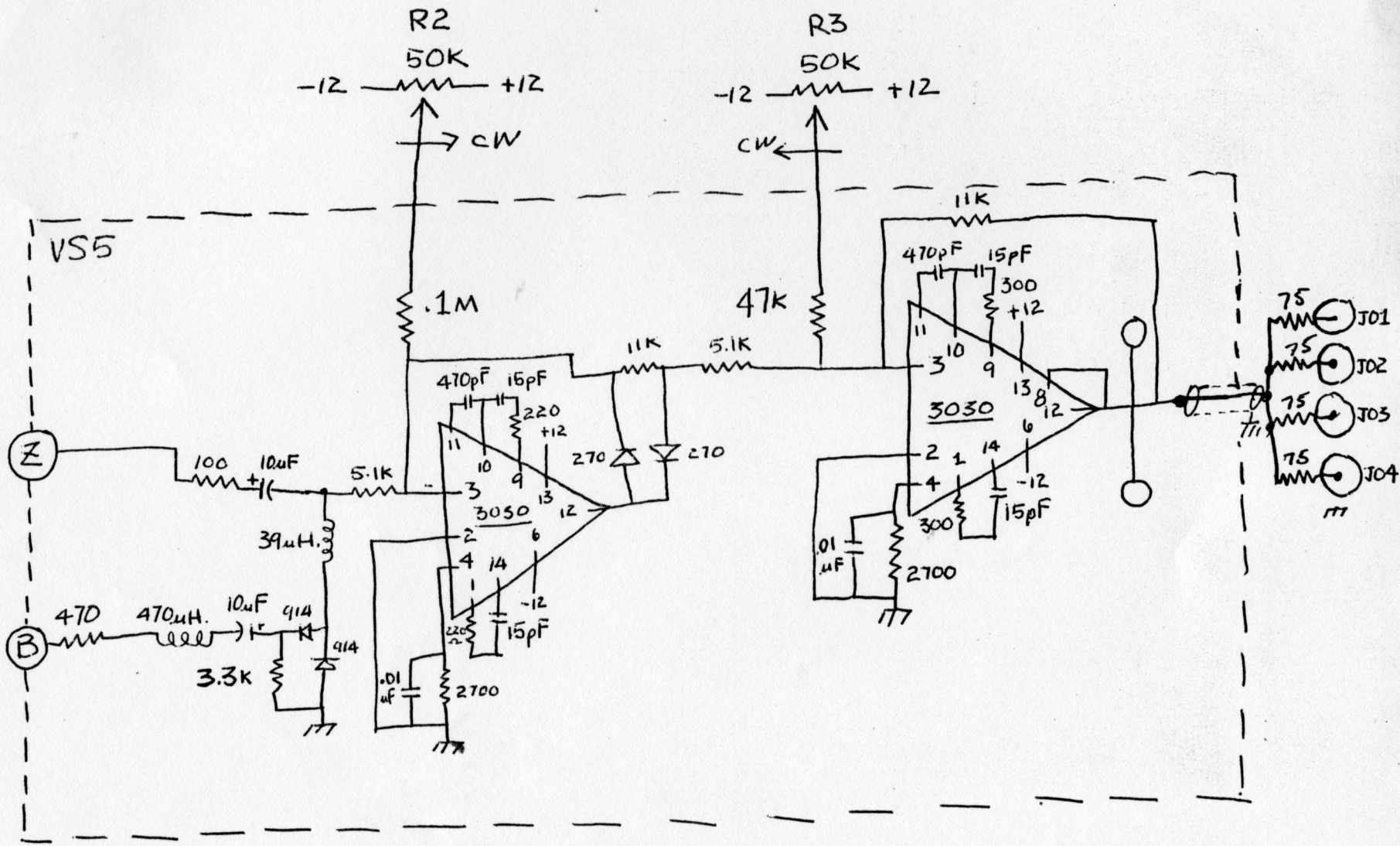

R1
R2
J01
J02
OK 9-75
J11
R3
J03
J04
R4
BURST FLAG
J05
J06
J07
J08
COMP. SYNC
RIT
R2T
HORZ. DRIVE
J09
J010
R3T
R4T
J011
J012
COMP. BLANK.
R5T
R6T
J013
J014
VERT. DRIVE
J019
J020
J021
J022
5/8"
1/8"
J015
J016
J017
J018
FRONT FACE
SYNC STRIPPER & CAMERA INPUT

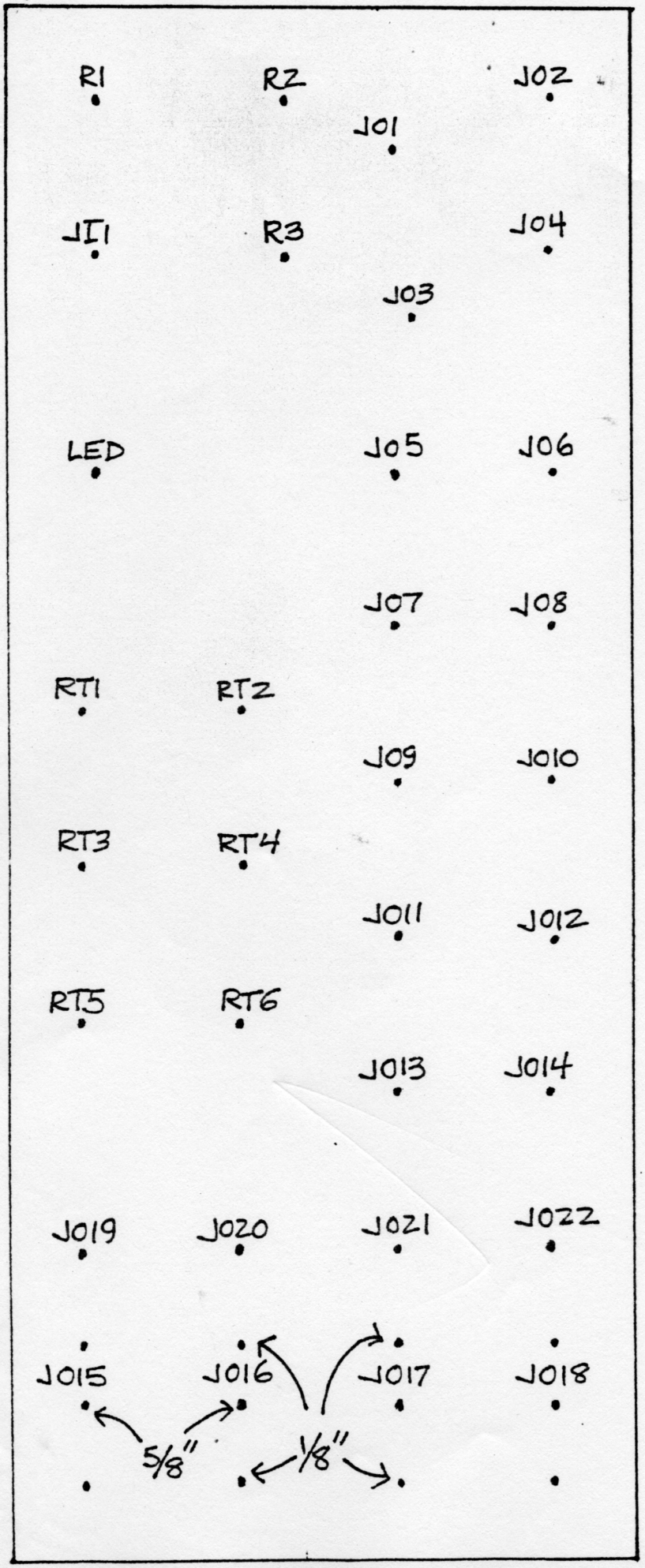


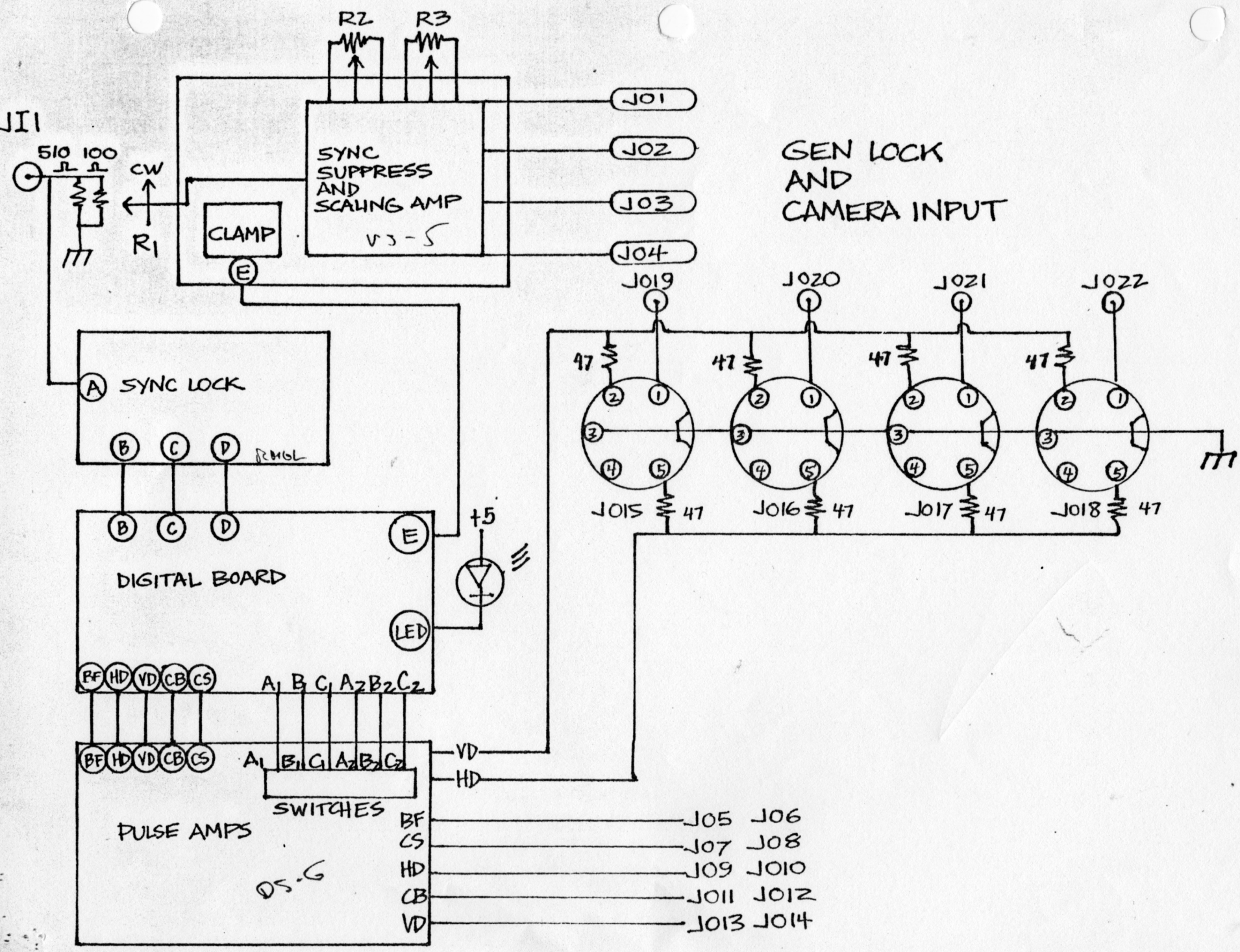
SYNC STRIPPER & CAMER[illegible]NPUT

SYNC SUPRESSOR AND SCALING AMP



SYNC STRIPPER & CAMERA INPUT


R1
R2
J02
J01
FRONT FACE
J11
R3
J04
J03
LED
J05
J06
BURST FLAG
J07
J08
COMP SYNC
RT1
RT2
J09
J010
HORIZ DRIVE
RT3
RT4
J011
J012
COMP BLANK
RT5
RT6
J013
J014
VERT DRIVE
J019
J020
J021
J022
GEN LOCK AND
CAMERA INPUT
J015
J016
J017
J018
5/8"
1/8"


R2
R3
J11
510
100
CW
R1
SYNC SUPPRESS AND SCALING AMP
CLAMP
J01
J02
J03
J04
GEN LOCK AND CAMERA INPUT
J019
J020
J021
J022
47
SYNC LOCK
A
B
C
D
E
DIGITAL BOARD
+5
LED
J015
J016
J017
J018
BF
HD
VD
CB
CS
A1 B1 C1 A2 B2 C2
SWITCHES
PULSE AMPS
J05 J06
J07 J08
J09 J010
J011 J012
J013 J014

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NORMOD
INSIDE
NOT NECESSARY

GEN LOCK AND CAMERA INPUT

This module locks its output pulses to an external black and white, or color video signal inputted at JI1. In addition, the signal at JI1 is clamped, sync suppressed, and available at JO1 - JO4. This latter process is identical to one third of the INPUT module, except that the clamp signal is generated internally (consult IP documentation for explanation).

JO5 - JO14 are various synchronization and drive pulses at -4 volts into 75 ohms.

JO19 - JO22 are video signal outputs from cameras connected to the EIA-J 6 pin connectors.

The front panel LED indicates when the gen-lock is locked to an external source. Light ON means that lock is present, light OFF means the module is not locked. Outputted sync is only stable for recording when the light is ON.

CIRCUIT DESCRIPTION

Video at JI1 goes to the Sync Lock board which contains a TBA920 integrated circuit. The TBA920 is a combination sync strip and horizontal phase locked loop oscillator. Stripped sync is converted to proper TTL digital voltage levels by one half of the 319 comparator. Sync is also filtered to pick off vertical. This resultant vertical trigger pulse is buffered by the other half of the 319, and used to initiate the vertical timing process on the digital board. The horizontal oscillator locks in both phase and frequency to an external video source. R1 controls the free running frequency (no video input) of the horizontal oscillator. R2 controls the phase of the oscillator when locked (video present). The leading edge of the horzontal oscillator pulse is usually set to coincide with the beginning of horizontal blanking.

The digital board has basically three sections. The horizontal timers (74123's) are driven by the horizontal oscillator pulses. They are set up so that all pulses during the horizontal blanking interval may be retimed and rephased. The vertical section is controlled by counters (74163's) and associated NAND gates and flip-flops. DIP switches A1, B1, C1, control the position of vertical blanking and drive. Switches A2, B2, C2, control the length of vertical blanking. Using A2 - C2, VB may be set anywhere from 17 to 24 horizontal lines long. Other

NAND gates on the board are used for sync recombination, and to detect when the device is locked properly to an external source.

SET UP

A dual trace oscilloscope is desirable, but not absolutely necessary. One vertical channel is connected to a high quality composite video signal (like the IP). The oscilloscope must be triggered from this source, either externally, or by channel selection. Set the scope to display several lines of video.

Follow the steps in the order given. Each step must be set up correctly before continuing to the next.

1. Using the other vertical scope channel, attach an oscilloscope probe to line B of the sync lock board. With NO VIDEO input adjust R1 so that the horizontal oscillator rate is approximately the same as that of the video.

2. With a video input at JI1, adjust R2 so that the phase of the horizontal oscillator pulse lines up with the beginning of horizontal blanking. This adjustment is a master horizontal phase control in that all pulses during the horizontal blaking interval are timed in relation to the leading edge of the horizontal oscillator pulse. When you change R2, all horizontal sync components will change phase accordingly.

3. Replace the scope probe with a 75 ohm line attached to composite blanking. Adjust RT5 for the proper length of horzontal blanking.

4. Looking at the output of horizontal drive, adjust RT6 for the proper length of HD.

5. Using the composite sync output, adjust RT3 for horizontal sync phase, and RT4 for sync length.

6. RT1 controls the phase position of burst flag, while RT2 controls its length. Burst flag is timed in relation to the leading edge of horizontal sync. Phase changes in sync will cause concurrent phase changes in burst flag.

7. Trigger the scope to display several fields of video. Using the vertical drive output, set DIP swithes A1,B1, and C1, to position VD at the beginning of the vertical blanking interval. The three switches have eight different possible position combinations. If VD is at best offset by one half line of video (advanced or delayed) from the beginning of vertical blanking, you are looking at the odd (B) field. In this case, set VD so that it is

advanced by one half line.

8. Attach the 75 ohm line to the composite blanking output. Vertical blanking should have the same phase as vertical drive. Adjust switches A2, B2, and C2, for the correct length of VB. 22 lines should be the maximum setting.

CONSTRUCTION

On the digital board, some wires are soldered from point to point, others are wire wrapped. Follow both the wire wrap list, and the pictorial. Remember to cut the foil, and jump ground to the outside bus, where noted.

All circuit boards, except the VS5 are mounted of 1" standoff mounting posts. VS5 is mounted at the top of the module, RMGL board in the middle, and the DS6 at the bottom. Mount the digital board behind the DS6, and RMGL boards. Input/output wires on the digital board should be kept to one edge, so that the board may be swung back for replacement of IC's, troubleshooting, etc.

Line E from the digital board runs to point JI2 on the VS5 board. Reverse the polarity of the 10uf cap. which is in series with the 470uh choke and the 1N914 diode.

On the Pulse Amp board (DS6), mount the 7808 voltage regulator to the heat sink, and then to the board itself with a 4x40 screw.

Remember to bus ground and the appropriate power to the circuit boards. +5 volts must also be run to the front panel for the LED, and the trimmers.

VS-5
DS-6-Pulse Amp
RMGL
DIGITAL